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MILITARY SPECIFICATION

MEATBALLS, SWEDISH, WITH BROWN GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers swedish meatballs with brown gravy thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. *Unless otherwise specified*, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Application for copies should be addressed to the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M. Street, S.W., Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SPICE AND TRADE ASSOCIATION (ASTA)

Official Analytical Methods of the American Spice and Trade Association

(Application for copies should be addressed to the American Spice and Trade Association, Inc., 580 Sylvan Avenue, Englewood Cliffs, NJ 07632.)

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and any of the following recognizable primal and sub-primal cuts: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, peeled knuckles, and bottom rounds. Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications (IMPS) cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used except tenderloins, rib fingers (intercostal meat), and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh-chilled state and shall be in excellent condition; i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration

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due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days after initial certification is permitted provided the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 120 days after initial certification is permitted provided that the beef is:

-frozen to 0°F or lower within 72 hours after initial certification
-stored at 0°F or lower
-protected from freezer deterioration and damage
-stored in containers that are adequate to maintain product excellence
-held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

3.2.1.3 Meatballs, cooked, frozen. Meatballs received in a cooked, frozen state shall be accompanied by a USDA Agricultural Marketing Service (AMS) Certificate certifying the requirements of 3.2.1, 3.2.1.1, 3.2.1.2, 3.3.1, and 3.3.2 have been met. The cooked meatballs shall have been vacuum packaged and heat sealed within 4 hours after the cooking process. The packaging material shall have an oxygen permeability rate of not more than 10 cc of oxygen per 24 hours at 21°C and 0 percent relative humidity. Within 72 hours after packaging, the packaged meatballs shall have been frozen to a temperature not greater than 0°F. The cooked, packaged frozen meatballs shall have been held at a temperature of 0°F or lower for not more than 30 days prior to tray pack filling.

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3.2.2 Allspice, ground. Ground allspice shall be derived from the dried, nearly ripe fruit of Pimenta officinalis Lindl and shall possess a fragrant, clove-like aroma, strongly aromatic, pungent, clovelike flavor and a dark reddish brown color. The allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with eugenol as a principal constituent. The allspice shall be of such size that not less than 95 percent passes through a U.S. Standard No. 25 sieve.

3.2.3 Bread crumbs. Bread crumbs shall be produced from white bread only. The bread crumbs shall be coarse ground, free flowing, and free from burnt, rancid, moldy, or musty odors or flavors.

3.2.4 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.5 Nutmeg, ground. The ground nutmeg shall be derived from dried kernel of Myristica fragrans Houtt and shall have a dull brown color. The nutmeg shall contain not less than 7.5 mL of volatile oil per 100 grams of ground nutmeg and shall be of such size that not less than 95 percent passes through a U.S. Standard No. 20 sieve.

3.2.6 Oregano, cracked or coarse ground. Oregano shall be derived from the dried leaves of the Origanum vulgare L. and shall possess a strong camphoraceous aroma and a pungent, slightly bitter flavor. The oregano shall contain not less than 2.0 mL of volatile oil per 100 grams of oregano and shall be of such size that not less than 95 percent passes through a U.S. Standard No. 20 sieve.

3.2.7 Paprika, ground. The ground paprika shall be Spanish paprika (Capsicum annum L.) and shall possess a bright orange to red color with an extractable color value of not less than 110 American Spice and Trade Association (ASTA) Color Units. The ground paprika shall be of such size that not less than 95 percent passes through a U.S. Standard No. 30 sieve.

3.2.8 Pepper, black, ground. Ground black pepper shall be derived from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic penetrating odor, a hot, biting, pungent flavor, and a light gray to speckled black-gray color. The ground pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground black pepper and shall be of such size that not less than 95 percent passes through a U.S. Standard No. 16 sieve.

3.2.9 Salt. Salt shall be non-iodized, white, refined sodium chloride, with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

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3.2.10 Sherry flavoring. Sherry flavoring shall be derived from natural sherry wine by removing the alcohol and excess water to produce an eight-fold concentrate.

3.2.11 Soup stock, dehydrated. Dehydrated soup stock shall be a light brown to tan colored free-flowing powder derived from soup stock produced from USDA inspected beef bones with adhering meat. The dehydrated soup stock shall have a minimum of 93 percent total solids and a maximum of 2 percent fat.

3.2.12 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized, modified maize starch for use in thermostabilized foods and shall comply with MIL-STD-900. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under sheer stress, and under conditions of cold storage. The cooked modified starch slurry shall be bland with essentially no cereal or starch taste.

3.2.13 Fat, frying. Frying fat shall be prepared from animal or vegetable (excluding coconut oil) fats, or a blend of animal and vegetable fats, which have been hydrogenated and deodorized. The frying fat shall have a clean bland flavor. The frying fat may be liquid or plasticized, may contain antioxidants in combinations and amounts allowed by FDA, and shall contain antifoaming agents. The fat shall have a maximum free fatty acid content of 0.05 percent (oleic), a stability of not less than 40 hours (A.O.M.), and a smoke point of not less than 400°F.

3.2.14 Water. Water used for ice making, formulation, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.15 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formulation.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef meeting the requirements of 3.2.1.1 shall be further trimmed, if necessary, so as to assure compliance with finished product requirements. The beef shall be coarse ground through a plate with holes measuring 1 inch in diameter. The ground beef shall be held for not more than 24 hours in the temperature range of 28° to 40°F prior to meatball preparation.

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3.3.2 Meatball preparation. The meatballs shall be formulated and prepared as follows:

Ingredient	Percent by weight
Beef	92.70
Bread crumbs	5.50
Sherry flavoring	1.00
Salt	0.75
Pepper, black, ground	0.03
Allspice, ground	0.01
Nutmeg, ground	0.01

a. The ingredients shall be combined and thoroughly mixed. The meatball mixture shall be ground through a 3/16-inch plate and formed into round meatballs of not less than 2-1/4 ounces nor more than 2-3/4 ounces each. The meatballs shall be maintained in the temperature range of 28° to 40°F for not more than 4 hours prior to the browning (cooking) process.

b. The meatballs shall be surface browned until no pink color is visible in the interior of the meatball. The meatballs may be cooked by any commercial method except moist heat (steaming) so as to conform to the finished product requirements. Frying fat, if used, shall comply with requirements of 3.2.13. The fat shall be filtered at least daily and the frying system cleaned at least weekly. The fat shall be discarded when foaming, undesirable odors and flavors or other evidence of fat breakdown is apparent. If the meatballs are not filled into the tray pack immediately after cooking, they shall be held for not more than 24 hours in the temperature range of 28°F to 40°F prior to tray pack filling.

c. The cooked meatballs may be stored in the frozen form provided within 4 hours after the cooking process, the meatballs are vacuum packaged and heat sealed in packaging material complying with the requirements specified in 3.2.1.3. The meatballs shall be frozen to 0°F or below within 72 hours after packaging. The packaged, frozen, cooked meatballs shall be maintained at 0°F or below for not more than 30 days prior to tray pack filling.

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3.3.3 Gravy preparation. The gravy shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	91.37
Starch, modified, high opacity	4.75
Soup stock, dehydrated	3.00
Salt 1/	0.75
Paprika	0.05
Pepper, black, ground	0.02
Oregano	0.02
Nutmeg	0.02
Garlic powder	0.02

1/ The total amount of salt in the gravy formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements.

a. The starch and part of the water shall be mixed to form a slurry.

b. The balance of ingredients shall be heated to a boil in a steam jacketed kettle with continuous and vigorous mixing to assure the spices are uniformly dispersed.

c. The steam supply to the kettle shall be shut off and the starch slurry added and uniformly mixed to form the gravy.

d. The volume of the gravy shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The gravy shall be maintained in the temperature range of 150° to 180°F for not more than 4 hours prior to filling into tray pack cans.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements.

a. A minimum of 24 meatballs shall be placed in a single layer within each tray pack can.

b. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirement (see 3.6s).

c. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

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3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material collectively, in any individual can shall be not more than 1 ounce.

e. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. Meatball texture shall not be dry, rubbery, tough, or mushy.

g. Meatballs shall be fairly round and firm enough to retain shape.

h. There shall be not less than 24 intact meatballs in each can.

i. Drained weight of 24 meatballs in a can shall be not less than 48.0 ounces.

j. The average drained weight shall be not less than 50.0 ounces.

k. The gravy shall not be lumpy.

l. The viscosity of the gravy shall be not less than 4.0 cm per 10 seconds nor greater than 18.0 cm per 10 seconds when determined by a Bostwick Consistometer, or not less than 20,000 centipoise nor greater than 110,000 centipoise when determined by a Brookfield Viscosimeter equipped with a number 4 spindle stirring at a speed of 1.0 RPM.

m. The average net weight shall be not less than 106 ounces.

n. No individual can shall have a net weight of less than 104 ounces.

o. The average fat content of the finished product shall be not greater than 12.0 percent.

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p. No individual tray pack shall have a fat content greater than 14.0 percent.

q. No individual tray pack shall have a salt content less than 0.5 percent or greater than 1.3 percent.

r. The product shall show no evidence of excessive heating (materially darkened or scorched).

s. Filled, sealed and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements of 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or not being in excellent condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	20	2	3	5	6
501 to 1200	32	3	4	7	8
1201 to 3200	50	5	6	10	11
3201 to 10000	80	7	8	14	15
10001 to 38000	125	10	11	21	22
Reinspection lots of 10001 to 38000	200	14	15	21	22

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
101		Presence of popliteal, prescapular, prefemoral, or any exposed lymph gland measuring 0.5 inch or more in any dimension.
102		Presence of blood clot measuring 1.0 inch or more in any dimension
	201	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
103		Presence of bruise measuring 1.0 inch or more in any dimension.
	202	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
104		Presence of bone measuring 0.3 inch or more in any dimension.
105		Presence of cartilage measuring 0.5 inch or more in any dimension.
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	203	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 2.0 square inches or more.
	204	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.

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TABLE II. Boned and trimmed beef defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	205	Presence of heavy connective tissue on lower edge of short plate or flank.
	206	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	207	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3.0 square inches or more.
108		Presence of knuckle cover.
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	208	Presence of kidney, pizzle eye, prepubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	210	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	211	Presence of dehydrated surface measuring 1.0 square inch or more.
	212	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	213	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension.

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TABLE II. Boned and trimmed beef defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	214	Presence of cod, udder, kidney, or pelvic fat.

1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall cause rejection of the involved lot.

4.5.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

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4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. Tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for 35 minutes in boiling water, opened, and inspected for the defects listed in table IV.

TABLE III. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	---	0	2
	8	16	1	2
3201 to 35000	13	---	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in a can is more than 1 ounce.
102	Presence of a bone piece measuring 0.3 inch or more in any dimension in an individual can.
103	Drained weight of 24 meatballs in a can is less than 48.0 ounces. 3/ 4/
104	Meatballs not fairly round or not firm enough to retain shape.
105	Texture of meatballs dry, rubbery, tough, or mushy.

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TABLE IV. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	
106	Can contains less than 24 intact meatballs.
107	Gravy is lumpy.
108	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off, strained through a U.S. Standard No. 8 sieve, and reserved for viscosity determinations. The remaining contents of the can shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and remove gravy without breaking the meatballs. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the meatballs. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. Twenty-four intact meatballs shall be weighed to determine the drained weight. Results shall be reported to the nearest 0.1 ounce.

4/ If the sample average drained weight is less than 50.0 ounces, the lot shall be rejected.

4.5.3.3 Fat and salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by an AMS agent as reserve samples.

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The three tray packs for laboratory analyses shall individually be tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products, except that preparation of the samples shall be as follows: The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The test results shall be reported to the nearest 0.1 percent. The lot shall be rejected if:

a. The average fat content of the three tray packs is greater than 12.0 percent.
b. The fat content of any individual tray pack is greater than 14.0 percent.
c. The salt content of any individual tray pack is less than 0.5 percent or greater than 1.3 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.3.4 Viscosity testing. The strained free liquid collected from each of the cans in the first sample of cans selected for drained weight inspection (see 4.5.3.2 and 3/ to table IV) shall be individually tested for viscosity as specified in 4.5.3.4.1 or, alternatively, as specified in 4.5.3.4.2 (see 6.4).

4.5.3.4.1 Bostwick Consistometer method.

Instrument: Bostwick Consistometer
Catalog Number: 23270-004
VWR Scientific Company
P.O. Box 7900
San Francisco, CA 94120

or

Catalog Number: 15-347-50
Fisher Scientific
585 Alpha Drive
Pittsburgh, PA 15238

Method:

a. Level the instrument.

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b. Bring gravy to 100° ± 1°F in a water bath in a covered container.

c. Stir gravy thoroughly before filling the Bostwick cavity.

d. Scrape gravy evenly across upper edge of cavity.

e. Release gravy and time gravy flow to the nearest 1 second and measure distance traveled to the nearest 0.1 cm.

The lot shall be rejected if the Bostwick viscosity value of the gravy from any can in the sample is less than 4.0 cm per 10 seconds or greater than 18.0 cm per 10 seconds.

4.5.3.4.2 Brookfield Viscosimeter method

Instrument: Brookfield Viscosimeter (variable speed) equipped with a No. 4 spindle.

Brookfield Engineering Lab, Inc.
Stoughton, MA 02072

Method:

a. Bring gravy to 80° ± 1°F in a water bath in a covered container.

b. Stir gravy thoroughly and pour 500 mL of the gravy into a 600 mL beaker.

c. Using a number 4 spindle, stir at a speed of 1.0 RPM and record dial reading.

d. Convert dial reading to centipoise. Record result to nearest 100 centipoise.

The lot shall be rejected if the Brookfield viscosity value of the gravy from any can in the sample is less than 20,000 centipoise or greater than 110,000 centipoise.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or of record of can seam teardowns shall be cause for rejection of any involved product.

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4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, only lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconforming tray pack can shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

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5.1.1.1 When specified (see 6.1), the base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 98 pounds per base box. The body design shall be reinforced on the bottom with six longitudinal beads or ribs 0.050 ± 0.010 inches deep. The beads shall be 9 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the bottom. The end design shall be reinforced with eight longitudinal beads 0.018 ± 0.004 inches deep. The beads shall be 11 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the end. The center four beads on the end design shall be interrupted to provide a 5-inch by 5-inch flat center panel for labeling. The apex of all these beads shall be outward. The interior and exterior can coating requirements shall be as specified in 5.1.1.

5.1.2 Level C. The product shall be preserved as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F load.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1 except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

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5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (e.g., EST 38) or a three letter code identifying the establishment.
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year. (Example, March 19, 1987 would be coded as 7078.) The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

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TO HEAT: Submerge unopened can in boiling water. Simmer gently for 40 to 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven for approximately 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 12 portions of 2 meatballs each.

5.5 Marking

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, when tray pack cans are fabricated from 90 pounds per base box, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. When tray pack cans are to be fabricated from 98 pounds per base box (see 5.1.1.1).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 The contracting officer may authorize an alternative contractor recommended method of viscosity testing if the alternative method is approved by U.S. Army Natick, Research, Development, and Engineering Center.

6.5 Subject term (key word) listing.

Beef
Canned foods
Meatballs
Rations
Thermostabilized
Tray pack

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-A543

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-44316	Meatballs, Swedish, with Brown Gravy, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 20 Nov 84

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